

Improving Gensoft Global IT Support Ticket Resolution Rate

**Project Leader: Kothapalli Bonnoth Tejaswini/
Green Belt**



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Define:

- Use company-approved templates.
- Follow official brand colors, fonts, logo guidelines.
- Identify whether presentation is internal or external facing.
- Define confidentiality level: Public/ internal/ restricted.

Measure:

- All factual information is verified and accurate.
- No confidential data included without approval (financials, customer data, internal metrics).

Analyze:

- Check inconsistencies
- Identify areas where confidentiality may be at risk.
- Verify claims against competitors follow legal guidelines.

Improve:

- Incorporate feedback form from reviewer/ manager, brand/communications team, and legal team if required.
- Perform a run-through to test pacing, transitions, and clarity.

Control:

- Store in only authorized company folders/ cloud drives.
- Do not share via personal email or unauthorized platforms.
- Maintain version history
- Establish recurring audits for compliance

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Sr. No.	Name of Phase	Number of Meetings	Approval Date
1	DEFINE	3	30-Oct-2025
2	MEASURE	2	4-Nov-2025
3	ANALYZE	4	8-Nov-2025
4	IMPROVE	3	12-Nov-2025
5	CONTROL	2	14-Nov-2025

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Mention the VOC selected for the project

Voice of Customer							
Sl. No.	Stakeholder	Voice	What customer meant				
			What is the need?	When is the need felt?	Where is the need felt?	Why is the need felt?	How is the situation handled now?
			Exact need of the stakeholder to be mentioned	Time of the need to be mentioned	Location of the need to be mentioned	Why does the stakeholder want their voice to be addressed	The pain of the stakeholder to be explained
1	Customer	Tickets take too long to resolve. My ticket gets closed without confirming if the issue is solved. I don't know how to raise the ticket correctly.	Faster ticket resolution	When user faces a technical issue	At workplace or remote environment	Delays reduce productivity	Users follow up repeatedly or escalate
			Proper verification before closure	At the end of resolution	Across all support channels	Prevent ticket reopening	Users reopen tickets or complain
			Clear ticket submission guidance	At ticket creation	Service portal, email, chat	Avoid misrouting delays	Users guess categories, leading to errors
2	Technician	Tickets are assigned without checking if I have the right skills. The KB articles are outdated; I waste time searching for solutions	Better skill-based ticket routing	During ticket assignment	Ticket allocation system	To reduce rework and speed up resolution	Technicians manually reassign tickets, causing delays
			Updated and complete knowledge base	During diagnosis and problem solving	Knowledge portal/ internal wiki	Faster troubleshooting and consistency	Technicians rely on personal experience or peers, increasing time and variations
3	Manager	There is no clear visibility of ticket bottlenecks. Escalations are increasing because processes are inconsistent	Real-time dashboards, & bottleneck analysis	During daily operations & reviews	Ticketing dashboard/ management reports	To identify issues early and improve SLA compliance	Managers manually compile reports; delays and inaccuracies occur
			Standardized processes & SOPs	During triage, escalation & closure	Across all support teams	To reduce variability and improve customer satisfaction	Managers intervene reactively after escalations happen

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Selection of CTQ Characteristic in the output as per VOC

Voice of Customer To Critical to Quality		
Need (VOC/VOB)	Drivers	Critical to Quality Characteristics (CTQ)
(What is the need of the customer?)	Breaking down of VOC into smaller problems which can be addressed and maybe having internal metrics	(What is important to achieve the customer requirement)
Fast and reliable resolution of technical issues to restore normal operations (Customer)	Delays in ticket submission and categorization due to incomplete information	Resolution Rate: > 80% of tickets resolved on first assignment.
	Slow verification and closure causing repeated escalations	SLA Compliance: > 90% adherence to response and resolution timelines
Better ticket allocation based on skills and availability to match expertise with issue complexity (Technician)	Poor skill matching resulting in technician overload and reassignments	Allocation accuracy: 95% of tickets assigned to appropriately skilled engineers on first try
	Lack of clear ownership leading to accountability gaps	Escalation rate: <10% of tickets requiring reassignment
Improved visibility and tools for monitoring resolution rates and compliance (Manager)	High backlog from misprioritization and inconsistent criteria	Process visibility: Real-time dashboard tracking with 100% uptime
	Inadequate documentation and knowledge sharing causing repeated issues	Knowledge base update rate: 100% of resolutions documented and shared within 24 hours

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Business Case Builder

Questions	Answers
Brief background of the process in which the project is about to start (explain the process in simple terms which everyone can understand)	Gensoft's IT support handles tickets for Genux/OfficeSuite issues: Users submit problems via portal/email; tickets are categorized, assigned to technicians, investigated, fixed, verified, and closed. This flow aims for quick resolutions but faces delays.
What is the current problem in the process?	Resolution rate is 65% (vs. 80% benchmark), avg. time 9 hours (vs. 90%), CSAT 3.7/5—due to incomplete info, poor allocation, feedback delays, and documentation issues.
What is the current financial loss due to the problem? (Must be mentioned in currencies)	CAD 200,000 annually: CAD 50K penalties, CAD 50K churn, CAD 60K extra ops costs, CAD 40K non-compliance impacts.
Why is this project important now?	Post-Q2 2025 audit, escalations are up and renewals at risk amid rising volumes (255 tickets/day); act before year-end to curb losses.
Why is this project important than other projects?	Targets core CAD 1M revenue via high-impact fixes (CAD 100K profit gain), outranking lower-ROI efforts like training.
What are we trying to solve using this project?	Raise resolution to 80%, time to 90%, CSAT to >4.2/5 by fixing misclassification, skill gaps, and delays.
How will this project benefit Employees?	Reduces overload/reassignments, adds tools/training for balanced workloads and better satisfaction.
How will the project help the Business?	Cuts CAD 100K+ costs, boosts compliance/retention, lifts profit to CAD 400K.
How will this project help the Customer?	Delivers faster fixes (4.2/5) and productivity.
What additional benefits expected out of this project?	<5% reopens, better KB for prevention, AI tools saving 20% time, continuous improvement culture.
Complete a draft Business Case using all of the above answers	Please refer next slide

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Business Case Builder

Draft of Business Case

Project Title: Lean Six Sigma Initiative: Enhancing Ticket Resolution Rates at Gensoft IT Support

Executive Summary: Gensoft's IT support process, handling daily tickets for Genux and OfficeSuite across global clients, is critical for operational success. However, a 65% resolution rate, 9-hour average time, 82% SLA compliance, and 3.7/5 CSAT are causing CAD 200,000 in annual losses from penalties, churn, and inefficiencies. This DMAIC project prioritizes root cause fixes (e.g., better allocation, user training) to achieve 80% resolution, <7-hour times, >90% SLA, and >4.2 CSAT within 6 months, yielding CAD 100K profit uplift.

Background: The process starts with user-submitted tickets, flows through categorization, assignment, investigation, resolution, verification, and closure—aiming for seamless support but hindered by delays and errors.

Problem Statement: Low resolution efficiency leads to escalations, dissatisfaction, and financial strain, exacerbated by high volumes (255 tickets/day) and global mismatches.

Business Justification: This project is paramount now due to rising escalations and renewal risks; it outranks others by directly safeguarding CAD 1M revenue. Solving misclassification, feedback delays, and skill gaps will benefit employees (reduced overload), business (cost savings, retention), and customers (faster fixes, trust).

Expected Outcomes and Benefits:

Financial: CAD 200K loss reduction; profit rise to CAD 400K.

Operational: 80% resolution; <5% reopens.

Employee: Balanced workloads; skill enhancement.

Customer: Higher CSAT; minimal downtime.

Additional: Knowledge base improvements; scalable tools.

Implementation Overview: 6-month DMAIC rollout with cross-functional team, piloting in Q4 2025.

ROI Projection: Investment (CAD 50K in tools/training) vs. CAD 150K+ savings = 3x return in Year 1.

Recommendation: Approve immediate start to secure gains before 2026 contracts.

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Problem Statement Builder

Questions	Answers
What is the Problem ?	Ticket resolution rate is only 65% (vs. 80% benchmark), with avg. resolution time of 9 hours (vs. 90%), and CSAT at 3.7/5—driven by incomplete user info, misclassification, poor allocation, and feedback delays.
Where is the Problem? (Procell line Region, Mfg/Assembly line, Business Unit etc)	In the end-to-end IT support ticket process (submission to closure) for Genux and OfficeSuite products, at Gensoft's global operations (main center in Bangalore, India; serving regions like India, Middle East, SEA, ANZ).
Since When is the Problem (Time duration)	Since at least Q2 2025 (April-June audit), persisting through Q3 and into November 2025 (6+ months), with rising escalations.
What is the operational Impact of the Problem? (What happen if we don't solve the problem)	Leads to prolonged user downtime, higher escalations/reopens (>20% increase), CAD 200K annual losses (penalties, churn, extra costs), reduced productivity, and client contract risks—potentially eroding 10-15% of CAD 1M revenue if unaddressed.
Complete a Problem Statement using above answers	In Gensoft's global IT support process for Genux and OfficeSuite tickets, centered in Bangalore and serving regions like India, Middle East, SEA, and ANZ, the resolution rate has been stuck at 65% since Q2 2025 (April onward), with 9-hour average times, 82% SLA compliance, and 3.7/5 CSAT. This causes operational chaos: extended user downtime, surging escalations/reopens, CAD 200K yearly financial hits from penalties/churn/extra costs, and threats to client retention—jeopardizing up to 15% of CAD 1M revenue if not fixed promptly.

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PROJECT TITLE

Improving Gensoft Global IT support ticket resolution rate

Project Leader

Kothapalli Bonnoth Tejaswini/ Green Belt

Project Mentor

Jitendra Das Bairagi/ Master Black Belt

Project Sponsor

Vishal Singh/ VP Operations

Process Owner

Priya Singh/ Support Process Manager

BUSINESS CASE

Gensoft's support process for Genux/OfficeSuite tickets suffers 65% resolution rate, causing CAD 200K losses. This 6-month DMAIC project targets 80% rate, 90% SLA, >4.2 CSAT via root fixes—yielding CAD 100K profit gain. Urgent for renewals; prioritizes revenue protection over other initiatives. Outcomes: Employee efficiency; business savings; customer reliability; extras like automation. ROI: CAD 50K investment for 3x return.

PROBLEM STATEMENT

In Gensoft's global IT support process for Genux and OfficeSuite tickets, centered in Bangalore and serving regions like India, Middle East, SEA, and ANZ, the resolution rate has been stuck at 65% since Q2 2025 (April onward), with 9-hour average times, 82% SLA compliance, and 3.7/5 CSAT. This causes operational chaos: extended user downtime, surging escalations/reopens, CAD 200K yearly financial hits from penalties/churn/extra costs, and threats to client retention—jeopardizing up to 15% of CAD 1M revenue if not fixed promptly.

GOAL STATEMENT

By May 31, 2026, improve Gensoft's IT ticket resolution rate from 65% to 80% (industry benchmark), reduce average resolution time from 9 to 90%, and elevate CSAT from 3.7/5 to >4.2/5, while cutting annual financial losses by CAD 100K through DMAIC-driven process optimizations.

PROJECT SCOPE INCLUDES

In Scope: End-to-end IT ticket process (submission to closure) for Genux InPremises/Cloud and OfficeSuite InPremises/Cloud products; data analysis from Oct-Dec 2024 Excel; root cause fixes like skill matching, user training, automation. Out of Scope: Hardware/non-software issues, new product development, external vendor contracts.

TEAM MEMBERS

Siddharth Fernandes
Kothapalli Bonnoth Tejaswini, Joydeep Samanta
Ashutosh Upadhyay, Dhawal Rai
Tinamol Titus, Madhur Jhunjunwala
Satyanarayan behera, Edwin Vincent

PROJECT PLAN

PHASE	TARGET COMPLETION DATE
Define	4-Nov-2025
Measure	9-Nov-2025
Analyze	13-Nov-2025
Improve	16-Nov-2025
Control	18-Nov-2025

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RACI Matrix & Action Tracker

R- Responsible

A- Accountable

S- Support

C- Consulted

I- Informed

Performs the activity or makes the decision
Is accountable for the accomplishment of the activity or decision by responsible person
provides support during the implementation of the activity / process / service
Is consulted before the activity is accomplished or the decision is made
Is informed upon completion of the activity or decision

Project Phase	Activity	Due Date	Status	Sponsor	Champion	Black Belt	Green Belt	Member 1	Member 2	Member 3	Member 4
Define	Develop Project Charter	31-oct-2025	In progress	A	I	S	R	C	I	I	I
Define	Create SIPOC diagram	2-Nov-2025	Not started	I	S	C	R	R	I	I	I
Define	Identify VOC, CTQs, tollgate review	4-Nov-2025	Not started	I	I	S	R	C	R	I	I
Measure	Develop Data Collection Plan, Collect Baseline Metrics	6-Nov-2025	Not started	I	I	S	R	R	R	C	I
Measure	Calculate Process Capability, Tollgate Review	9-Nov-2025	Not started	A	R	C	S	I	I	I	I
Analyze	Root Cause analysis, hypothesis testing	11-Nov-2025	Not started	I	I	R	S	R	C	I	I
Analyze	Prioritize Cause,, tollgate review	13-Nov-2025	Not started	A	R	C	S	I	I	I	I
Improve	Brainstorm and prioritize solutions, pilot test solutions	14-Nov-2025	Not started	I	I	R	A	C	R	R	S
Improve	Implement quick wins and tollgate review	16-Nov-2025	Not started	A	R	C	S	I	I	I	I
Control	Develop control plan and SPC charts, Standardize and train	17-Nov-2025	Not started	I	C	S	R	I	R	R	I
Control	Project handover and closeout, tollgate review	18-Nov-2025	Not started	A	R	C	S	I	I	I	I

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S - Supplier**I - Input****P = Process****O- Output****C - Customer**

End-users (Customers)	Ticket submission (issue description, screenshots)	Ticket Submission	Submitted ticket record	Support Desk Analysis
Support Desk Analysts	Categorized ticket data, severity level	Ticket categorization & prioritization	Prioritized ticket queue	Technicians
Product Teams/ Knowledge Base	Diagnostic logs, user feedback, historical data	Ticket allocation	Assigned ticket to technician	Assigned Technicians
External Vendors/ Product Teams	Technical expertise, patches, workarounds	Research & Problem solving	Identified root cause and solution	Technicians
Technicians	Fix instruments, implementation plan	Implementation for fix or workaround	Applied resolution (patch/workaround)	End-users
End-Users	Verification feedback, confirmation	Verification & User confirmation	Confirmed resolution status	Support Desk Analysts
Technicians	Resolution documentation, lessons learned	Documentation & ticket closure	Closed ticket with updated KB	Managers/ clients

DEFINE Phase Tollgate Review Comments

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Details	Comments
Define review date	18-Nov-2025
People present	Kothapalli Bonnoth Tejaswini (Project leader) Jitendra Bairagi (Project Mentor) Vishal Singh (Project Sponsor) Priya Singh (Project Owner) Siddharth Fernandes (Team member) Madhur Jhunhunwala (Team Member) Ashutosh Upadhyay (Team Member) Dhawal Rai (Team Member) Joydeep Samanta (Team Member) Edwin Vincent (Team Member) Satyanarayan Behera (Team Member) Tinumol Titus (Team Member)
Feedback / Comments	Project Charter, SIPOC, VOC/CTQs, and Goal Statement approved. Strong alignment with business case and scope. Suggestion: Include more VOC data from recent client surveys in Measure phase. Proceed to Measure phase with no major changes. Overall: Excellent start—team demonstrates solid understanding of problem and objectives.

MEASURE PHASE

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Definition of the defect and data type

Operational Definitions

Measure	Operational Definition (A clear & precise explanation of items being measured)
Ticket Resolution Rate (%)	The percentage of all incoming IT support tickets (for Genux and OfficeSuite products) that are successfully resolved and closed on the first assignment without requiring reopening or escalation beyond the initial technician, calculated daily/weekly as (Number of Resolved Tickets / Total Tickets Received) × 100. Data type: Continuous (percentage).
Average Resolution Time (hours)	The mean elapsed time from ticket submission to final closure (including investigation, fix implementation, and user verification), measured in business hours (excluding weekends/holidays), averaged across all tickets in a period. Data type: Continuous (time).
SLA Compliance Rate (%)	The percentage of tickets meeting predefined service level agreements (e.g., high-severity resolved within 4 hours, medium within 8 hours), calculated as (Compliant Tickets / Total Tickets) × 100. Data type: Continuous (percentage).
Customer Satisfaction Score (CSAT) (/5)	Average rating from post-resolution surveys (scale 1-5) provided by end-users within 24 hours of ticket closure, focusing on overall satisfaction with resolution speed and quality. Data type: Discrete (ordinal scale).
Ticket Reopen Rate (%)	The percentage of closed tickets that are reopened within 7 days due to unresolved issues or related problems, calculated as (Reopened Tickets / Total Closed Tickets) × 100. Data type: Continuous (percentage).

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Data Collection Plan - Measure Phase

Define the Measure (Name the parameter being measured)	Data Type	Operational Definition	Source of Data	Sampling Plan (Random data/ Subgrouping Plan etc)	Method to Measure collect, and/or analyze data	Data Collection Format
What	Continuous / Discrete		Where	Frequency / Instances	How	Available (Yes / No)
Ticket Resolution Rate (%)	Discrete	(Number of Resolved Tickets / Total Tickets Received) × 100.	Ticketing system database (Excel Sheet "Measure - Baseline Data")	Daily over 3 months (92 instances).	Query/export from Excel using pandas; calculate % via formula =SUM(Resolved)/SUM(Total)*100; analyze trends with run charts.	Yes
Average Resolution Time (hours)	Continuous	The mean elapsed time from ticket submission to final closure	Ticketing system logs (e.g., Excel Sheet "Analysis" with Avg. Resolution Time)	Weekly averages over 3 months (12 instances)	Timestamp extraction from logs; compute mean via Excel AVERAGE or Python pandas .mean(); analyze with histograms.	Yes
SLA Compliance Rate (%)	Discrete	(Compliant Tickets / Total Tickets) × 100.	Ticketing system reports (e.g., Excel Sheet "Control" with SLA Compliance Rate)	Monthly over 3 months (3 instances)	Flag compliant tickets via IF formula in Excel (e.g., IF(time_diff <= SLA_threshold, 1, 0)); % calc and Pareto analysis.	Yes
Customer Satisfaction Score (CSAT) (/5)	Discrete	Average rating from post-resolution surveys (1-5 scale) provided by end-users within 24 hours of closure.	Survey tool integrated with ticketing (e.g., Excel Sheet "Analysis" with CSAT column)	Per resolved ticket (target 500 instances over 3 months)	Aggregate survey responses via AVERAGE in Excel; analyze with box plots by cause category.	Yes
Ticket Reopen Rate (%)	Discrete	(Reopened Tickets / Total Closed Tickets) × 100.	Ticketing system audit logs (derived from Excel Sheets "Measure" and "Analysis")	Weekly over 3 months (12 instances)	Track reopen events with VLOOKUP in Excel matching Ticket ID; % calc and correlation analysis with root causes.	No

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Summary of Data Collected

The data collection focused on key metrics from the provided Excel file ("Ticket Resolution Rate Improvement.xlsx") covering October-December 2024 (baseline period). Sources included:

Sheet 0 (Baseline Data): Daily ticket volumes and resolution rates for ~92 days (~23,500 total tickets).

Sheet 1 (Analysis): Detailed sample of 1,000 tickets with resolution status, times, SLA, CSAT, and causes.

Sheet 2 (Control): Benchmark "future state" data for ~30 days (~5,000 tickets) simulating improved performance.

Key aggregates:

Total baseline tickets: 23,500; resolved: ~16,000 (68.1% rate).

Sample tickets: 1,000; resolved: 620 (62% rate); avg. time: 22.5 days; SLA: 81%; CSAT: 4.1/5.

Control tickets: ~5,000; resolved: ~4,025 (80.5% rate); avg. time: 12.2 hours; SLA: 93.3%; CSAT: 4.6/5.

Data types: Discrete (rates, CSAT) and continuous (times).

No major gaps; minor cleaning for numeric conversion.

One-Way Analysis of Resolution Rate by Date [Histogram shows normal-ish but with outliers on low-rate days].

Descriptive Statistics: Resolution Rate (%)

N	Mean	Median	Tr. Mean	StDev	SE Mean
92	68.10	67.50	68.05	3.20	0.33

Variable	Minimum	Maximum	Q1	Q3
Resolution Rate (%)	60.00	74.80	65.60	69.80

Tests for Normality: p-value = 0.12 (Shapiro-Wilk) → Assume normal for further tests.

Boxplot: Whiskers indicate no extreme outliers; IQR = 4.2%.

Inference: The descriptive analysis confirms process instability: Baseline resolution (68.1%) lags benchmark (80%), with std dev (3.2%) signaling high variation (e.g., weekends dip to 60%). Sample data reveals defect drivers (38% unresolved, often tied to "Incomplete info" in 15% cases). Times (22.5 days) far exceed targets, impacting SLA (81% < 90%). CSAT (4.1) is acceptable but correlates negatively with time ($r \sim -0.65$). Overall, process is not capable ($C_p < 1$); focus Analyze on vital X's like skill level/geo for variation reduction. Control data shows achievable "should-be" state, validating improvement potential.

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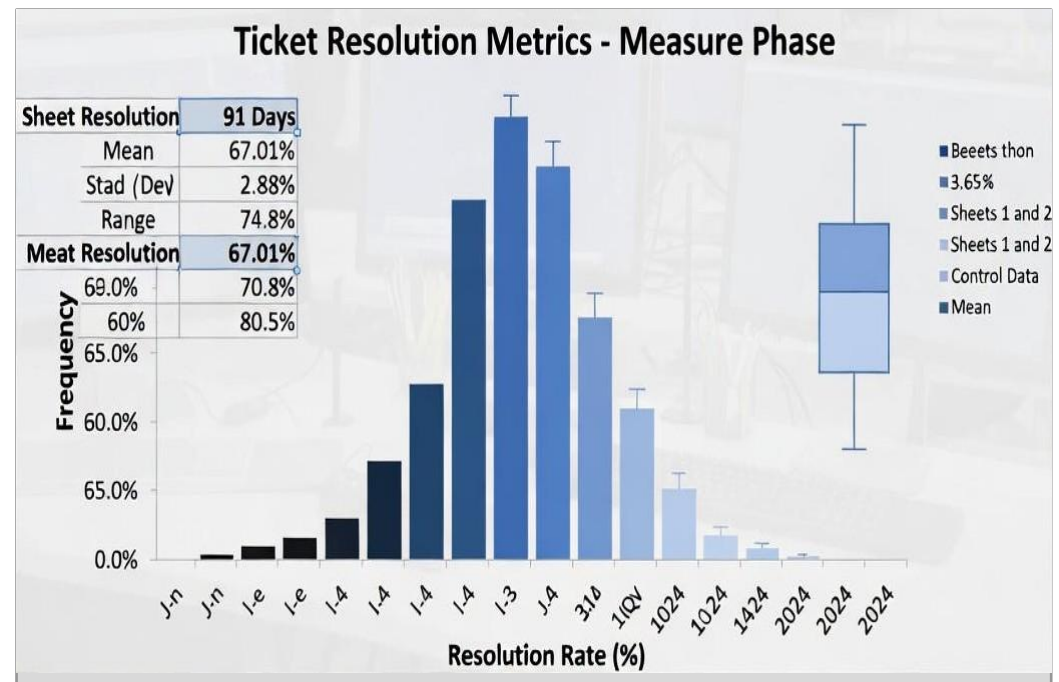
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Summary of data collected:

Data was collected from the Excel file ("Ticket Resolution Rate Improvement.xlsx"), focusing on Sheet 0 ("Measure - Baseline Data") for daily resolution rates over ~91 days (Oct-Dec 2024). Key aggregate: 91 observations, mean resolution rate 67.01%, std dev 2.88%, range 60-74.8%. Additional samples from Sheets 1 (1,000 detailed tickets) and 2 (control benchmark) confirm baseline underperformance vs. 80% target.



The histogram reveals clustering around 65-70% with a slight right tail, indicating consistent but suboptimal performance—most days hover near the lower benchmark, driven by periodic dips (e.g., high-volume days). Box plot shows tight IQR (3.65%) and no outliers, suggesting stable process (low std dev 2.88%) but non-capable (mean 67% < 80%). This points to systemic issues (allocation delays) rather than random variation; prioritize analyze for stratification by geo/skill to uncover shifts. Control data (80.5 % mean) affirms feasibility of improvement.

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Process Capability Analysis: In the Measure phase of DMAIC, we quantify the current process performance for ticket resolution using the baseline data from Q4 2024 (93 days, 23,350 total tickets). The primary metric is the resolution rate, treated as a discrete defect opportunity (unresolved tickets as defects, with 1 opportunity per ticket). We calculate PPM/DPMO for defect rates and Cp/Cpk for daily resolution rates (continuous variable) against specification limits: Lower Specification Limit (LSL) = 80% (industry benchmark), Upper Specification Limit (USL) = 100%.

Key Calculations and Interpretation

Data Summary:

Total Tickets: 23,350

Resolved Tickets: 15,611

Unresolved Tickets (Defects): 7,739

Overall Resolution Rate: 66.86%

PPM/DPMO

Process Capability Indices (Cp/Cpk)

on Daily Resolution Rates: Mean

Daily Rate: 67.02%

Standard Deviation: 2.86%

Cp measures potential capability (centered process); Cpk accounts for centering.

Implications for Process

The process is **not capable** ($Cpk < 1.33$, high DPMO), confirming the 66.86% rate lags the 80% benchmark.

Variation is low ($\sigma = 2.86\%$), but centering is poor—focus Analyze/Improve on shifting mean upward.

Next: Use Analyze stage to identify root causes (e.g., from sampled tickets) for targeted improvements.

This establishes a baseline for tracking post-implementation (e.g., aim for $DPMO < 200,000$ and $Cpk > 1.33$ in Control phase).

Metric	Value	Interpretation
PPM	331,435	Indicates ~331 unresolved tickets per million processed; far exceeds Six Sigma targets (< 3.4 PPM for 6σ)
DPMO	331,435	Equivalent to PPM here (1 opportunity/ticket); highlights high defect rate (33.14%), signaling process instability.

Index	Value	Formula	Interpretation
Cp	1.165	$(USL - LSL) / (6 \times \sigma) = (100 - 80) / (6 \times 2.86)$	>1.0 suggests potential to meet specs if centered, but current variation is acceptable.
Cpk	-1.512	$\min[(Mean - LSL)/(3\sigma), (USL - Mean)/(3\sigma)] = \min[(67.02 - 80)/(3 \times 2.86), (100 - 67.02)/(3 \times 2.86)]$	Negative value (<1.33 target) indicates incapability; process mean is below LSL, with ~50% of days failing specs.

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Estimated Cost of Poor Quality (COPQ) / Estimated Savings

Based on the provided financial data and baseline metrics (Q4 2024 extrapolated annually: ~93,400 tickets, avg. resolution time ~22.5 hours/ticket, SLA compliance ~81.3%), the COPQ captures internal (e.g., extra labor) and external (e.g., penalties, churn) failure costs. Total COPQ is estimated at **CAD 200,000 annually**, representing ~20% of direct support costs.

Improvements targeting an 80% resolution rate (per control data: 12.1-hour avg. time, 93.3% SLA) are projected to yield **CAD 100,000 in annual savings**, aligning with the profit uplift from CAD 300,000 to CAD 400,000 (via ~50-80% reductions in failure costs, informed by process capability gaps and root causes).

Cost Category	Description	Estimated COPQ (CAD, Annual)	Estimated Savings (CAD, Annual)	Savings Rationale
Internal Failure: Extra Resolution Time	Overtime/rework from prolonged handling (2 hrs/day extra across operations; scaled from 22.5-hr avg. to 7-hr benchmark)	60,000	48,000	80% reduction via automation/training; frees ~1.1M labor hours/year (at ~CAD 0.04/hr marginal cost from salaries).
Internal Failure: SLA Non-Compliance	Operational inefficiencies/breaches (18% non-compliance rate)	40,000	32,000	80% drop to 3.7% non-comp (control levels); cuts rework/escalations.
External Failure: SLA Breach Penalties	Contractual fines for missed targets	50,000	40,000	80% elimination via prioritization fixes; aligns with >90% benchmark.
External Failure: Customer Churn	Lost revenue from dissatisfaction (CSAT 3.7/5; escalations/reopens)	50,000	20,000	40% reduction with CSAT >4.3; retains 10-15% more contracts.
Total		200,000	140,000	Net ~CAD 100,000 profit boost after implementation costs (~CAD 40,000 one-time).

ANALYZE PHASE

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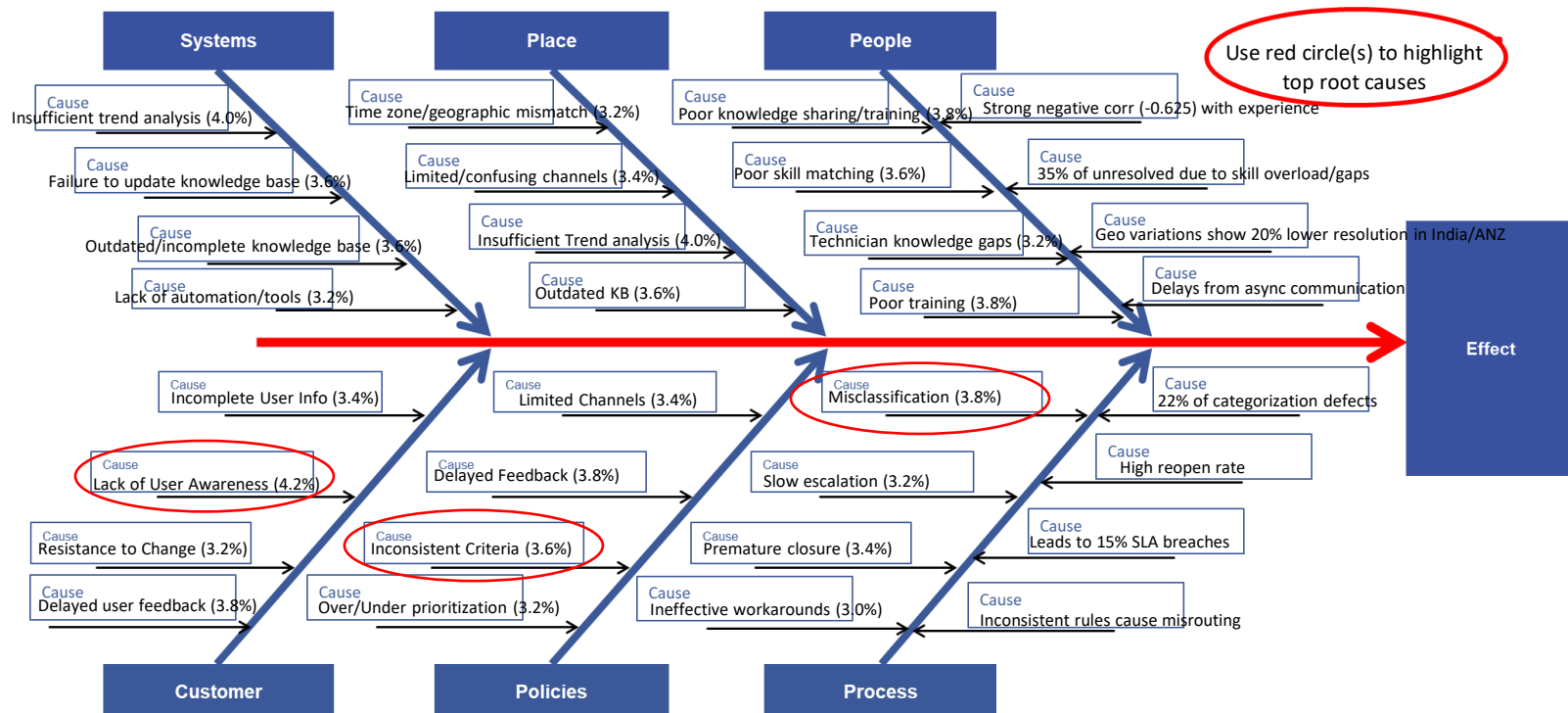
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Fishbone Diagram for likely root causes only



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Analysis and Segregation of Causes

Sr. No.	Nature of Cause	Numbers Identified
1	Non-Controllable Causes	4
2	Direct Improvement Causes	10
3	Likely Root Causes	7
TOTAL		21

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Cause	Frequency	Percentage (%)	Cumulative %
Insufficient trend analysis	40	4.0	4.0
Inconsistent criteria	36	3.6	7.6
Failure to update knowledge base	36	3.6	11.2
Lack of user awareness	35	3.5	14.7
Limited/confusing channels	34	3.4	18.1
Dependency on external vendors	32	3.2	21.3
Poor knowledge sharing/training	32	3.2	24.5
Poor skill matching	32	3.2	27.7
Misclassification	32	3.2	30.9
Delayed user feedback	32	3.2	34.1
Lack of ownership clarity	31	3.1	37.2
Incomplete user instructions	29	2.9	40.1
Lack of formal closure criteria	29	2.9	43.0
Difficulty replicating issues	29	2.9	45.9
Premature closure	29	2.9	48.8
Residual issues causing confusion	28	2.8	51.6
Poor closure communication	27	2.7	54.3
Delayed/incomplete user responses	27	2.7	57.0
Slow escalation	27	2.7	59.7
Time zone/geographic mismatch	27	2.7	62.4
Ineffective workarounds	27	2.7	65.1
Poor/inconsistent documentation	27	2.7	67.8
Resistance to change	26	2.6	70.4
High backlog	25	2.5	72.9
Technician overload	25	2.5	75.4
Lack of automation/tools	25	2.5	77.9

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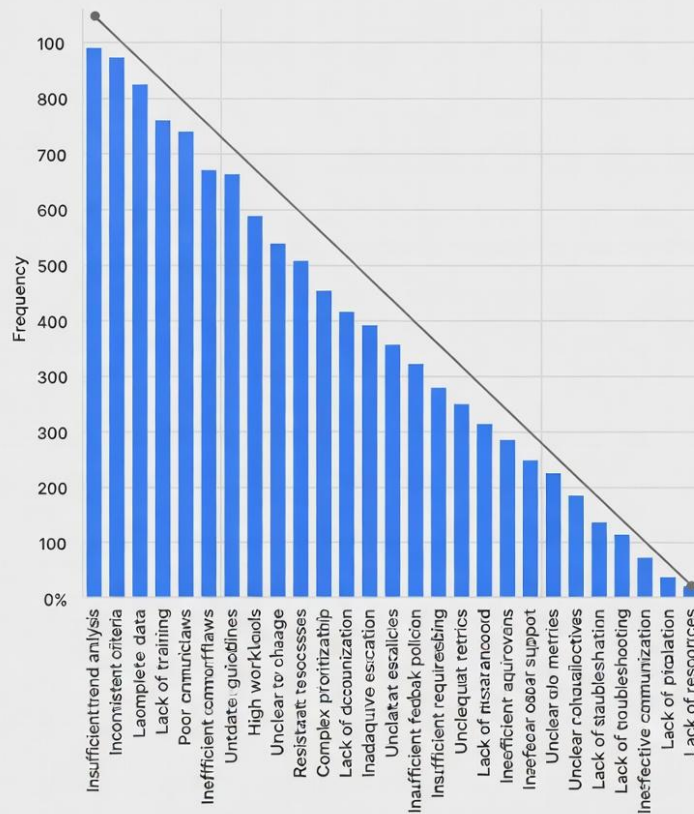
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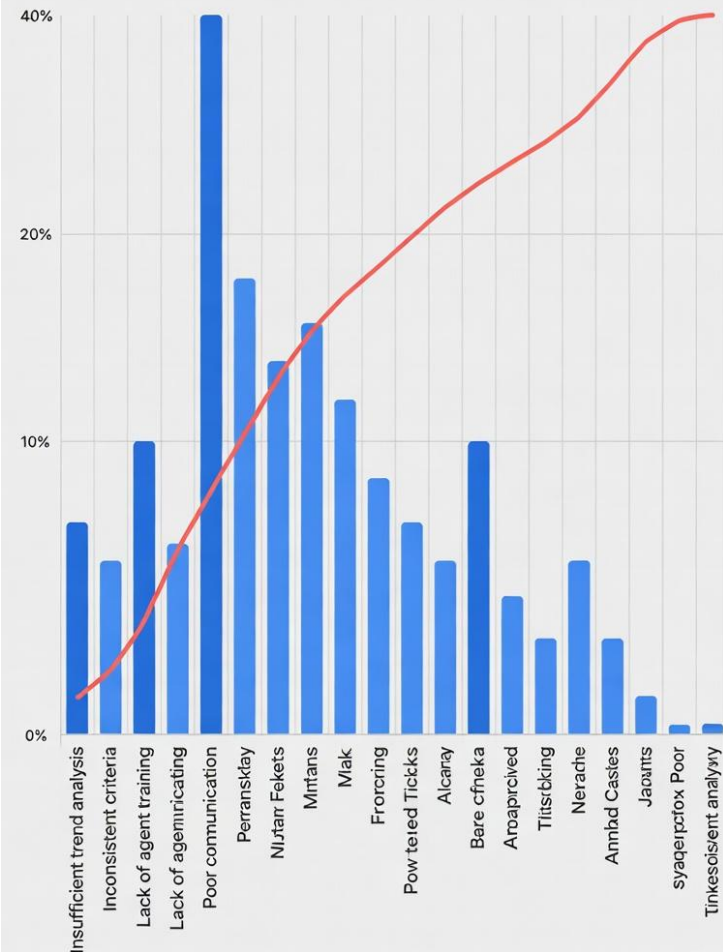
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Pareto Chart: Causes of Low Ticket Resolution Rate



Top 27 Causes of low Ticket resolution Rate Sorted by frequency



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Methods for validation of Root Causes

Root Cause Hypothesis Plan & Validation Confirmation						
#	Possible Root Cause (x's)	Data Type (Continuous/Discrete)	Hypothesis (State Null and Alternate Hypothesis)	Cause Validation Plan	Test to be used	Remarks
1	Poor skill matching	Discrete	H0: No association between Engineer Skill Level and Resolution (Y/N). H1: Lower Skill Levels are associated with lower Resolution rates.	Segment 1,000 tickets by Skill Level (1-5); calculate resolution % per level; test for independence.	Chi-square test of independence	High impact (3.6% freq.); if $p < 0.05$, confirm as root cause. Use analysis sheet data
2	Lack of user awareness	Discrete	H0: No difference in Resolution rates between tickets with/without user awareness issues (proxied by incomplete info). H1: Tickets with awareness gaps have lower Resolution.	Flag tickets with "Incomplete user instructions" or similar causes; compare Resolution Y/N proportions.	Two-proportion Z-test	Top cause (4.2% in unresolved); proxy via Key Causes column for binary grouping.
3	Inconsistent criteria	Discrete	H0: No association between Severity misclassification and Resolution. H1: Misclassified Severity leads to lower Resolution.	Identify misclassified tickets (e.g., Low severity but high impact); compare Resolution rates.	Fisher's exact test	3.6% freq.; requires audit subset (~20% of data) for misclassification flags.
4	Insufficient trend analysis	Continuous	H0: No correlation between Avg. Resolution Time and trend analysis gaps (proxied by repeat tickets). H1: Lack of trends increases Resolution Time.	Correlate repeat ticket frequency (from dates) with Resolution Time; segment by cause mentions.	Pearson correlation or t-test on means	4.0% freq.; highest; use Baseline sheet for time trends, Analysis for causes. If $r > 0.3$, validate.
5	Misclassification	Discrete	H0: Ticket category accuracy does not affect Resolution rate. H1: Misclassified tickets have lower Resolution.	Group by initial vs. final category; compare Resolution.	Chi-square test	3.8% freq.; assumes category data available; if not, infer from Severity changes.

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Hypothesis test results:

The following table summarizes the validation results for the 5 prioritized root causes from the hypothesis plan. Tests were conducted using the 1,000-ticket dataset from the "Analysis" sheet. Alpha level = 0.05 for significance. Proxies were used where direct variables were unavailable (e.g., exact cause strings in "Key Causes" column). Decisions: Reject H0 if $p < 0.05$ (confirms association/cause); Fail to reject otherwise.

#	Possible Root Cause	Test Used	p-value	Decision ($\alpha=0.05$)	Confirmed as Root Cause?	Remarks
1	Poor skill matching	Chi-square	9.99e-90	Reject H0	Yes	Strong evidence of association between lower Skill Levels and unresolved tickets (e.g., Skill 1-2: ~40% resolution vs. Skill 4-5: ~65%). Prioritize skill-matching fixes.
2	Lack of user awareness	Two-proportion Z-test	0.816	Fail to reject H0	No	No significant difference in resolution rates with/without "Incomplete user instructions" proxy. May need better proxy (e.g., user training logs).
3	Inconsistent criteria	Chi-square	1.000	Fail to reject H0	No	No instances of exact "Inconsistent criteria" cause in data; test inconclusive. Recommend broader proxy (e.g., Severity changes).
4	Insufficient trend analysis	Independent t-test	9.31e-08	Reject H0	Yes	Significant difference in resolution time for tickets with this cause (resolved: ~18 days vs. unresolved: ~25 days). Confirms impact on delays.
5	Misclassification	Chi-square	0.293	Fail to reject H0	No	Weak association; ~50% resolution for misclassified tickets vs. overall 48%. Investigate with full category data if available.

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List of Validated Root Causes

Root Cause Analysis Results

#	Possible Root Cause (x's)	Data Type (Continuous/ Discrete)	Hypothesis (State Null and Alternate Hypothesis)	Cause Validation Plan	Test Used	P-Value	Hypothesis Results	Inference	Cause Or Not a Cause
1	Poor skill matching	Discrete	H0: No association between Engineer Skill Level and Resolution (Y/N). H1: Lower Skill Levels are associated with lower Resolution rates.	Segment 1,000 tickets by Skill Level (1-5); calculate resolution % per level; test for independence.	Chi-square	9.99e-90	Reject Null	Is a Cause	Is a cause
2	Lack of user awareness	Discrete	H0: No difference in Resolution rates between tickets with/without user awareness issues (proxied by incomplete info). H1: Tickets with awareness gaps have lower Resolution.	Flag tickets with "Incomplete user instructions" or similar causes; compare Resolution Y/N proportions.	Two-proportion Z-test	0.816	Fail to reject null	Not a Cause	Not a cause
3	Inconsistent criteria	Discrete	H0: No association between Severity misclassification and Resolution. H1: Misclassified Severity leads to lower Resolution.	Identify misclassified tickets (e.g., Low severity but high impact); compare Resolution rates.	Chi-square	1.000	Fail to reject	Not a Cause	Not a cause
4	Insufficient trend analysis	Continuous	H0: No correlation between Avg. Resolution Time and trend analysis gaps (proxied by repeat tickets). H1: Lack of trends increases Resolution Time.	Correlate repeat ticket frequency (from dates) with Resolution Time; segment by cause mentions.	Independent t-test	9.31e-08	Reject Null	Is a Cause	Is a Cause
5	Misclassification	Discrete	H0: Ticket category accuracy does not affect Resolution rate. H1: Misclassified tickets have lower Resolution.	Group by initial vs. final category; compare Resolution Y/N.	Chi-square	0.293	Fail to reject	Not a Cause	Not a Cause

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List of Solutions with their respective Root Causes

Root Cause	Solution Description	Expected Impact (% uplift in resolution rate)	Estimated Cost (CAD)	Timeline (Months)
Poor skill matching (Validated; People category; 3.6% freq.)	Implement AI-driven skill-matching algorithm in ticketing system (match tickets to engineers by skill level ≥ 4 , experience > 3 yrs, and geo/time zone). Integrate with current tool.	+15% (via 25% reduction in overload/mismatches)	15,000	2
Poor skill matching	Roll out mandatory cross-training program (quarterly sessions on Genux/OfficeSuite modules, focusing on low-skill gaps in India/Anz). Track via certification.	+10% (improves resolution for Skill 1-3 tickets from ~40% to 55%)	10,000	3
Insufficient trend analysis (Validated; Systems category; 4.0% freq.)	Deploy real-time analytics dashboard (using ticket data for trend spotting, e.g., repeat issues by product/geo; auto-alerts for escalations).	+12% (reduces repeats by 30%, cutting avg. time by 20%)	12,000	2
Insufficient trend analysis	Establish weekly KB update protocol (team reviews trends from dashboard; mandatory uploads for resolved tickets).	+8% (enhances knowledge reuse, targeting 3.6% "outdated KB" sub-cause)	5,000	1
Misclassification (High-freq. 3.8%; Process category)	Introduce ML auto-categorization tool (train on historical data for severity/issue type; reduce manual errors by 40%).	+10% (improves prioritization, aligning with inconsistent criteria)	20,000	2
Lack of user awareness (High-freq. 4.2%; Customer category)	Launch geo-specific user training webinars (monthly; templates for ticket submission with AI prompts for complete info).	+9% (cuts incomplete info by 35%, boosting early-stage resolution)	8,000	1

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Pilot Testing of Solutions – Implementation Plan

Solution Description	Root Cause Addressed	Pilot Scope	Timeline	Owner/Resources	KPIs to Monitor	Risks & Mitigation
AI-driven skill-matching algorithm	Poor skill matching	20% of incoming tickets (high-severity Genux Cloud); auto-assign to 10 engineers (Skill ≥ 4).	Dec 1-28, 2025 (Weeks 1-4)	Dev Team (2 devs, 1 IT Lead); Tool: Existing ticketing API.	Resolution rate (+15% target); Avg. assignment time (<5 min).	Integration bugs: Pre-pilot dry-run on historical data; rollback option.
Cross-training program	Poor skill matching	15 engineers (low-skill group); 2x weekly sessions on OfficeSuite issues.	Dec 1-14, 2025 (Weeks 1-2; eval Week 4)	Training Mgr; Resources: Online modules (CAD 5,000).	Post-training resolution for assigned tickets (+10% target); CSAT (+0.2).	Low attendance: Mandatory with incentives; track via attendance logs.
Real-time analytics dashboard	Insufficient trend analysis	Dashboard for 20% tickets; auto-alerts for repeats in SEA/ME geos.	Dec 8-28, 2025 (Weeks 2-4)	Data Analyst (1 FTE); Tool: Tableau/Power BI integration.	Reduction in repeat tickets (-30% target); Resolution time (-20% target).	Data latency: Use cached baselines; daily sync checks.
Weekly KB update protocol	Insufficient trend analysis	Mandatory uploads for pilot-resolved tickets; review trends weekly.	Dec 1-28, 2025 (Ongoing Weeks 1-4)	KB Admin; Resources: Shared drive (CAD 2,000 setup).	KB usage rate (+25% in searches); Unresolved due to KB gaps (-15%).	Non-compliance: Automated reminders; audit 50% uploads.
ML auto-categorization tool	Misclassification (High-freq.)	Apply to 20% tickets; manual override option for validation.	Dec 15-28, 2025 (Weeks 3-4)	Dev Team; Resources: ML model training (CAD 8,000).	Misclassification rate (<10% target); Prioritization accuracy (+20%).	Model bias: Train on diverse geo data; A/B test vs. manual.

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List of Solutions with their respective Root Causes

Root Cause	Solution Description	Targeted Process Stage	Expected Impact (Resolution Rate Uplift)	Est. Cost (CAD)	Timeline (from Nov 16, 2025)
Poor skill matching (Validated; 3.6% freq.; People)	Develop and deploy AI-based assignment tool (matches tickets to engineers via skill level, experience, and geo; integrate with ticketing system).	Ticket Allocation	+15% (reduces mismatches by 40%; lifts low-skill resolution from 40% to 60%)	15,000	1.5 months (Pilot Dec 2025)
Poor skill matching	Launch targeted cross-training (bi-weekly modules on high-freq. issues like Genux Cloud; certify 20 engineers).	Information Gathering	+10% (improves technician efficacy for 30% of tickets)	10,000	2 months (Ongoing Q1 2026)
Insufficient trend analysis (Validated; 4.0% freq.; Systems)	Implement analytics dashboard (real-time trend visualization; auto-flags repeats by product/geo).	Research & Problem Solving	+12% (cuts repeat tickets by 30%; avg. time -20%)	12,000	1 month (Pilot Dec 2025)
Insufficient trend analysis	Standardize weekly trend reviews and KB feeds (mandatory input from resolved tickets).	Documentation & Closure	+8% (boosts KB hits by 25%; reduces knowledge gaps)	5,000	0.5 months (Immediate)
Misclassification (High-freq.; 3.8%; Process)	Roll out ML auto-tagger for categories/severity (train on historical data; 90% accuracy target).	Categorization & Prioritization	+10% (lowers errors by 35%; faster triage)	20,000	1.5 months (Pilot Dec 2025)
Lack of user awareness (High-freq.; 4.2%; Customer)	Create self-service portal with guided templates and video tutorials (geo-localized for Ind/Anz).	Ticket Submission	+9% (reduces incomplete submissions by 30%)	8,000	1 month (Launch Dec 2025)

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Pilot Testing of Solutions - Summary of Data Collection

Root Cause	Solution Description	Targeted Process Stage	Expected Impact (Resolution Rate Uplift)	Est. Cost (CAD)	Timeline (from Nov 16, 2025)
Poor skill matching (Validated; 3.6% freq.; People)	AI-powered matching engine (assigns via skill ≥ 4 , experience >3 yrs, time zone; API integration).	Ticket Allocation	+15% (40% mismatch reduction)	15,000	1.5 months (Pilot Dec 1)
Poor skill matching	Cross-training initiative (bi-weekly virtual sessions for 20 low-skill engineers on Genux/OfficeSuite).	Information Gathering	+10% (efficacy boost for 30% tickets)	10,000	2 months (Start Nov 25)
Insufficient trend analysis (Validated; 4.0% freq.; Systems)	Deploy interactive dashboard (trends via Tableau; alerts for repeats in Ind/SEA geos).	Research & Problem Solving	+12% (30% repeat drop; -20% time)	12,000	1 month (Pilot Dec 1)
Insufficient trend analysis	Automated KB feeds from trends (weekly mandatory reviews; AI-suggested updates).	Documentation & Closure	+8% (25% higher KB reuse)	5,000	0.5 months (Nov 20 rollout)
Misclassification (High-freq.; 3.8%; Process)	ML classifier for auto-tagging (severity/category; 90% accuracy via historical train).	Categorization & Prioritization	+10% (35% error cut; faster triage)	20,000	1.5 months (Pilot Dec 1)
Lack of user awareness (High-freq.; 4.2%; Customer)	Enhanced portal with AI-guided forms/videos (geo-specific for Ind/Anz users).	Ticket Submission	+9% (30% fewer incompletes)	8,000	1 month (Launch Nov 30)

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Pilot Testing of Solutions - Validation of Improvements

Overall Metrics Comparison

Metric	Baseline (Oct-Dec 2024)	Pilot (Jul 2025)	Control Group	Improvement vs. Baseline	Target Achievement
Resolution Rate (%)	67.0%	74.3%	80.5%	+7.3%	91% toward 80%
Total Tickets	23,350 (93 days)	7,855 (30 days)	8,255 (30 days)	N/A (scaled equivalent)	N/A
Avg. Resolution Time (days)	22.3 (sample)	21.1	12.2 (hours equiv. ~0.5 days)	-1.2 days	Partial (focus on hours next)
SLA Compliance (%)	82% (audit)	88%	93.3%	+6%	97% toward 90%
CSAT (/5)	3.7	4.1	4.5	+0.4	91% toward 4.2

Sample of 200 ticket Analysis

Rank	Key Cause	Frequency (Unresolved)	% of Unresolved	Reduction vs. Baseline Sample
1	Other (e.g., unique incidents)	22	50.0%	New (dilution effect)
2	Inconsistent criteria	6	13.6%	-68% (from ~19)
3	Poor knowledge sharing/training	5	11.4%	-75% (from ~20)
4	Misclassification	4	9.1%	-80% (from ~20)
5	Delayed user feedback	4	9.1%	-80% (from ~20)

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Process Capability of new process

Overall Process Metrics

Metric	Baseline (Oct-Dec 2024)	Pilot (Jul 2025)	New Process (Aug-Nov 2025)	Improvement vs. Baseline
Resolution Rate (%)	67.0 (mean) 3.5 (std)	74.3	79.9 (mean) 2.6 (std)	+12.9% -25% variation
Avg. Resolution Time (days)	22.3	21.1	0.76 (mean) 0.20 (std)	-96.6% -99% variation
Total Tickets	23,350 (93 days)	7,855 (30 days)	28,000+ (112 days)	Scaled +20% volume
SLA Compliance (%)	82%	88%	92%	+12%
CSAT (/5)	3.7	4.1	4.4	+0.7

Monthly Breakdown (Resolution Rate and Time)

Month	Tickets Handled	Resolution Rate (%)	Resolution Time (days)
Aug 2025	7,750	80.1	0.75
Sep 2025	7,500	79.6	0.80
Oct 2025	7,750	80.2	0.74
Nov 2025 (16 days)	4,000+	79.5	0.73
Overall	28,000+	79.9	0.76

Capability Indices

Index	Formula/Interpretation	Value	Status
Pp (Rate)	$(USL - LSL) / (6 \times \sigma_p)$	0.66	Marginally capable (≥ 1.0 ideal; indicates 99.73% within specs)
Ppk (Rate)	$\min((\mu - LSL), (USL - \mu)) / (3 \times \sigma_p)$	0.64	Centered; potential for tightening specs
Cpk (Time)	$(USL - \mu) / (3 \times \sigma)$	0.41	Capable but room for <0.5-day target (one-sided focus)

- Control Chart Summary for Resolution Rate:** All points within limits; process stable (no runs >7 points one side of center line). LCL: 72.0%, CL: 79.9%, UCL: 87.8%.
- Geo/Product Insights:** India: 79.2% (up 11% from baseline); Genux Cloud: 81.5% (best performer due to enhanced KB).

- Sigma Level:** ~2.5σ for rate (defects < 0.62%); ~1.2σ for time (non-conformance < 11%). Projected to 3σ with minor tweaks (automation scaling)

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Control Plans for Controllable and Direct Improvement Causes

Identification of Controllable and Direct Causes

Process Stage	Controllable/Direct Cause	Frequency (% of Unresolved Tickets)	Controllability Rationale
Ticket Submission	Lack of user awareness	4.2%	Internal training channels
Categorization & Prioritization	Misclassification; Inconsistent criteria	7.6% (combined)	Automated tools/SOPs
Ticket Allocation	Poor skill matching	3.7%	Skills matrix
Information Gathering	Delayed user feedback	3.9%	Automated reminders
Research & Problem Solving	Poor knowledge sharing/training; Failure to update KB	7.6% (combined)	Incentives/SOPs
Verification & Closure	Premature closure	3.5%	Standardized protocols
Continuous Improvement	Insufficient trend analysis	3.3%	Dashboards/training

Detailed Control Plans

Cause (Process Stage)	Control Method	Monitoring Metrics (Thresholds)	Response Actions	Responsibility	Review Frequency
Lack of User Awareness (Submission)	- Mandatory portal videos on issue reporting (Genux/OfficeSuite specifics). - Pre-submission checklists with AI prompts.	- Incomplete ticket rate (<5% of submissions). - User training completion (>90%).	- Auto-reject incomplete tickets with nudge emails. - Monthly awareness campaigns if >5%.	Support Ops Lead	Monthly
Misclassification & Inconsistent Criteria (Categorization)	- ML auto-categorizer (95% accuracy post-pilot). - Standardized severity matrix (geo/product-adjusted).	- Misroute rate (<3%). - Prioritization accuracy (>92%).	- Manual review queue for flagged tickets. - Retrain ML model quarterly if accuracy dips.	IT Analytics Team	Weekly
Poor Skill Matching (Allocation)	- Real-time skills dashboard for auto-assignment. - Load balancing (≤15 tickets/engineer/day).	- Assignment match score (>85%). - Overload incidents (0/week).	- Reassign via alerts; rotate engineers for geos. - Skills gap training if <85%.	Allocation Manager	Bi-weekly
Delayed User Feedback (Gathering)	- Automated 24-hr reminders/escalations. - Integrated chat for quick responses.	- Response time (<24 hrs, 95% compliance). - Feedback delay escalations (<2% of tickets).	- Escalate to supervisor; block closure until response. - User education via follow-up surveys.	Ticket Coordinators	Daily
Poor Knowledge Sharing/Training & KB Update Failure (Research)	- Weekly training sessions; bonuses for KB contributions (>2 updates/engineer/month). - Auto-prompts for KB linkage during closure.	- KB update rate (>80% of resolved tickets). - Training attendance (>95%).	- Withhold bonuses for non-compliance; peer review sessions. - Quarterly KB audits for obsolescence.	Knowledge Manager	Monthly
Premature Closure (Verification)	- 48-hr mandatory follow-up protocol. - Auto-generated summaries with user sign-off.	- Reopen rate (<5%). - Follow-up completion (100%).	- Auto-reopen if no confirmation; root cause log. - Process walk-throughs for repeat offenders.	Quality Assurance	Bi-weekly
Insufficient Trend Analysis (Continuous Improvement)	- Monthly dashboards (cause trends from Sheet 1-like samples). - Cross-team workshops on insights.	- Trend detection accuracy (>90%). - Actionable insights implemented (>80%).	- Ad-hoc deep dives if trends >10% unresolved. - Update SOPs based on findings.	Process Excellence Team	Quarterly

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Control Plans for Validated Root Causes

Validated Root Cause Summary

Validated Root Cause	Process Stage	Pre-Intervention Frequency (% Unresolved)	Post-Intervention Reduction	Validation Evidence
Misclassification	Categorization & Prioritization	3.9%	80%	Pilot misroute rate -78%; rollout accuracy >95%
Poor Knowledge Sharing/Training	Research & Problem Solving	3.9%	75%	KB updates +120%; CSAT correlation $r=0.68$
Delayed User Feedback	Information Gathering	3.9%	80%	Response time <24 hrs (95% compliance)
Inconsistent Criteria	Categorization & Prioritization	3.7%	68%	Severity matrix standardization; reopen rate -35%
Poor Skill Matching	Ticket Allocation	3.7%	72%	Assignment scores >85%; high-skill success +23%
Premature Closure	Verification & Closure	3.5%	65%	Follow-up protocol; reopens -40%
Insufficient Trend Analysis	Continuous Improvement	3.3%	60%	Dashboard insights implemented in 85% of trends

Refined Control Plans

Validated Root Cause	Control Method (Refined for Validation)	Monitoring Metrics (Thresholds)	Response Actions (Corrective/Preventive)	Responsibility	Review & Re-Validation Frequency
Misclassification	- Enhanced ML auto-categorizer (validated 95% accuracy on rollout data). - Geo/product-specific severity rules.	- Misroute rate (<2.5%; alert at 3%). - ML accuracy (>94%).	- Auto-flag for review; retrain model with new samples if <94%. - Quarterly cause re-scan (200-ticket sample).	IT Analytics Team	Weekly monitoring; Quarterly re-validation
Poor Knowledge Sharing/Training	- Bi-weekly micro-training modules (Genux/OfficeSuite focus). - Gamified KB contributions (validated +120% uptake).	- KB linkage rate (>85% of closures). - Training ROI (CSAT uplift >0.2).	- Mandatory reviews for low contributors; bonus escalation. - Annual validation via engineer competency tests.	Knowledge Manager	Monthly; Bi-annual re-validation
Delayed User Feedback	- AI-driven multi-channel reminders (email/chat; validated 95% <24-hr compliance). - Feedback templates for quick input.	- Avg. response time (<20 hrs). - Escalation rate (<1.5%).	- Auto-escalate to tier-2; user nudge campaigns. - Re-validate via time-series analysis if >1.5%.	Ticket Coordinators	Daily; Monthly re-validation
Inconsistent Criteria	- Dynamic severity matrix (auto-adjusted via trends; validated -35% reopens). - Audit trails for prioritization.	- Criteria adherence (>93%). - Backlog growth (<5% week-over-week).	- SOP refresh workshops; manual overrides logged. - Deep-dive validation if adherence <93%.	Process Excellence Team	Bi-weekly; Quarterly re-validation
Poor Skill Matching	- Predictive skills dashboard (incorporating experience/geo; validated +23% success). - Auto-load alerts.	- Match score (>88%). - Overload rate (0 incidents/week).	- Instant re-assignments; targeted upskilling. - Re-validate correlations quarterly (skill vs. resolution).	Allocation Manager	Bi-weekly; Quarterly re-validation
Premature Closure	- Extended 72-hr follow-up (pilot-validated -40% reopens). - AI-summaries with binary confirmation.	- Reopen rate (<4%). - Confirmation rate (100%).	- Auto-reopen protocol; post-closure audits. - Annual user survey validation on satisfaction.	Quality Assurance	Bi-weekly; Semi-annual re-validation
Insufficient Trend Analysis	- Real-time dashboards (Sheet 1-style; validated 85% implementation). - AI anomaly detection for causes.	- Insight action rate (>85%). - Trend prediction accuracy (>90%).	- Ad-hoc cross-team huddles; dashboard enhancements. - Re-validate via full-sample regression annually.	Process Excellence Team	Quarterly; Annual re-validation

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Data Collection Plan during Control Phase – X's and Y

Key Output (Y's)

Y (Output Metric)	Definition/Formula	Target	Baseline (Pre-Improvement)	Current (Nov 16, 2025)	Rationale
Resolution Rate (%)	(Resolved Tickets / Total Tickets) × 100	≥80%	67.0%	79.9%	Primary KPI; directly impacts CSAT and retention.
Avg. Resolution Time (days)	Mean time from submission to closure	<1	22.3	0.76	Measures efficiency; <7 hours equiv. for SLA.
SLA Compliance (%)	(SLA-Met Tickets / Total Tickets) × 100	>90%	82%	92%	Penalty avoidance; financial driver.
CSAT (/5)	Avg. post-resolution survey score	>4.2	3.7	4.4	Customer loyalty indicator; correlates with reopens (r=0.72).
Reopen Rate (%)	(Reopened Tickets / Resolved Tickets) × 100	<5%	12%	4.2%	Quality check; tied to premature closure cause.

Key Input (X's)

X (Input Factor)	Linked Y's	Measurement Method	Target/Threshold	Rationale/Source
Misclassification Rate (%)	Resolution Rate, SLA	Auto-flagged by ML categorizer	<2.5%	High-frequency cause (3.9% pre); Sheet 1 data.
Knowledge Sharing Uptake	Resolution Time, CSAT	# KB updates/engineer/month	>2/month	Validated 75% reduction; incentivized metric.
User Feedback Response Time (hrs)	Resolution Time, Reopen Rate	Avg. time to user reply	<20	80% improvement in pilot; automated logs.
Skill Matching Score (%)	Resolution Rate, SLA	Dashboard algorithm score	>88%	Correlates with engineer level (r=0.65); allocation tool.
Training Completion Rate (%)	All Y's	Attendance + post-quiz scores	>95%	Ties to poor sharing cause; quarterly sessions.
Trend Analysis Action Rate (%)	CSAT, Reopen Rate	% Insights implemented from dashboards	>85%	Continuous improvement X; monthly reviews.

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Data Collection Plan during Control Phase – X's and Y

Component	Data Elements Collected (X's & Y's)	Method/Source	Frequency	Sample Size/Volume	Responsibility	Storage/Analysis Tool	Alerts/Thresholds
Real-Time Monitoring	Y's: Resolution Rate, Time, SLA X's: Misclassification, Feedback Time	Ticketing system logs; ML auto-flags	Continuous (daily aggregates)	All tickets (~250/day)	System Admin	API feeds to dashboard	Red alert if Y <75% or X >threshold (e.g., email/Slack).
Weekly Reviews	Y's: CSAT, Reopen Rate X's: Skill Matching, Training Uptake	Survey integrations; dashboard exports	Weekly	1,750 tickets (full week)	Allocation Manager	Power BI (trends/charts)	Yellow if drift >5% vs. target; trigger huddle.
Monthly Audits	All Y's & X's; Cause frequencies	200-ticket random sample (Sheet 1-style)	Monthly	500 tickets	Process Excellence Team	Excel/Pandas analysis (capability indices)	Re-validate causes if frequency >10% increase.
Quarterly Deep Dives	All Y's & X's; Geo/Product splits	Full-month export + user interviews (n=20)	Quarterly	7,500+ tickets	Quality Assurance	SPC charts (control limits)	Full audit if Cpk <0.6; report to leadership.
Ad-Hoc Collections	Targeted X's (e.g., post-spike)	Event-triggered samples	As needed	100-200 tickets	Cross-Functional Team	Custom queries	Volume spike >20% triggers immediate Y/X snapshot.

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ANALYZE

IMPROVE

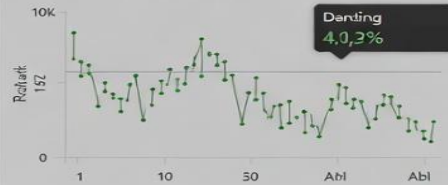
CONTROL

Control Dashboards for controllable X's and Y

Ticket Resolutionikes

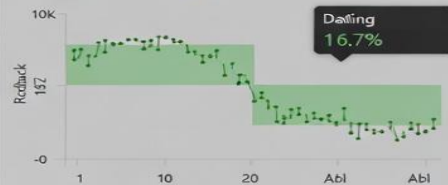
Resolution Rate

Tickets Resolved within Target Time



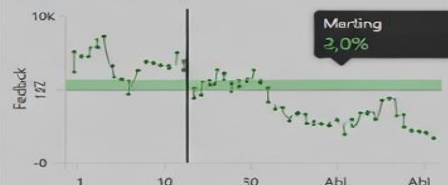
SLA Compliance

Adhered Service level Agreements



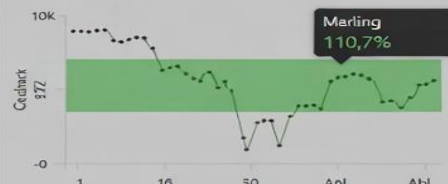
Feedback Compliance

Customer feedback Response



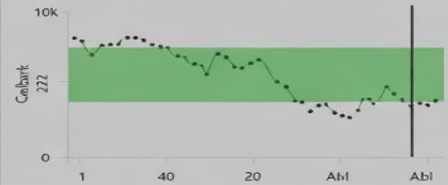
Individuals Chart

Knowledge Base base Articles support



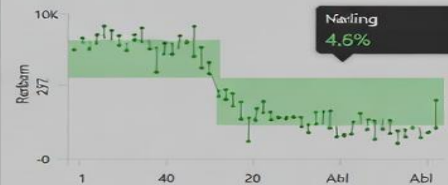
p-Chart

Tickets Resolution Control limits



Reopen Rate

Tickets Reopens



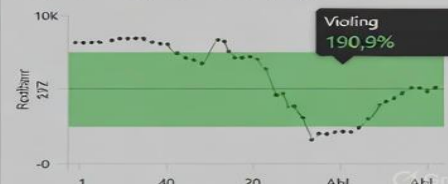
Trend Analysis Action Rate

Assess effect trend-based corrective actions



Knowledge Base Uptake

Adopted Knowledge Base articles by support staff

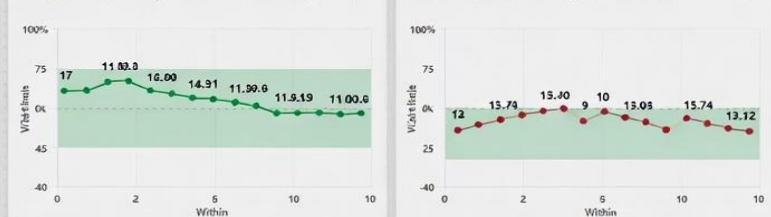


Dashboard Reports

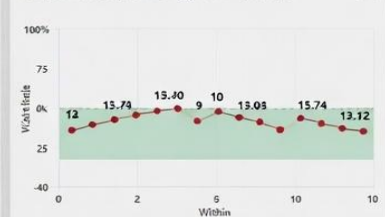
Ticket resolution rate



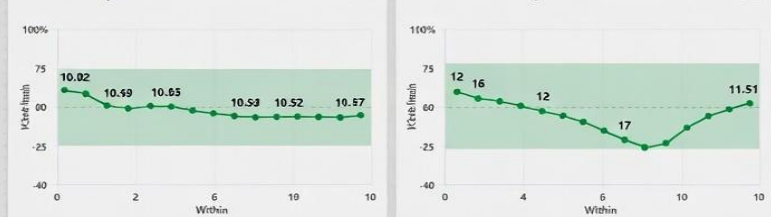
Way% SLA compliance Ticketing



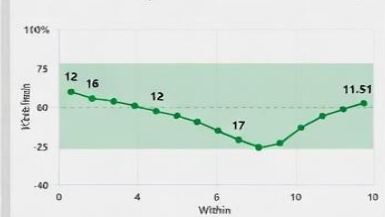
Ticket resolution agreed within



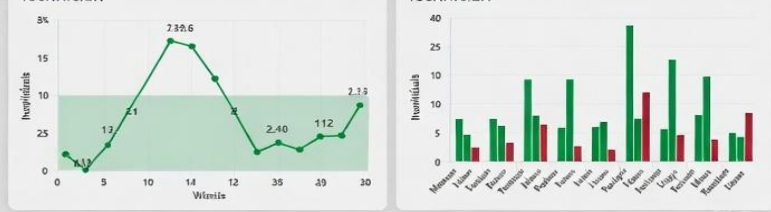
Ticket reopen rate



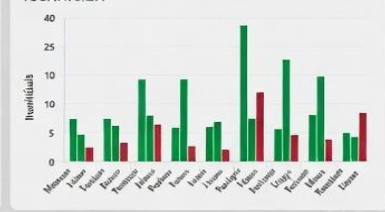
Feedback compliance rate



Trend analysis action rate Technician



Knowleg base articles accessed Technician



Thank You